DEDICATED INVERTER & STORAGE FIELD MANUAL

New Energy Grid

# Enerlution Hybrid Inverters & $LiFePO_4$ Battery Storage Guide

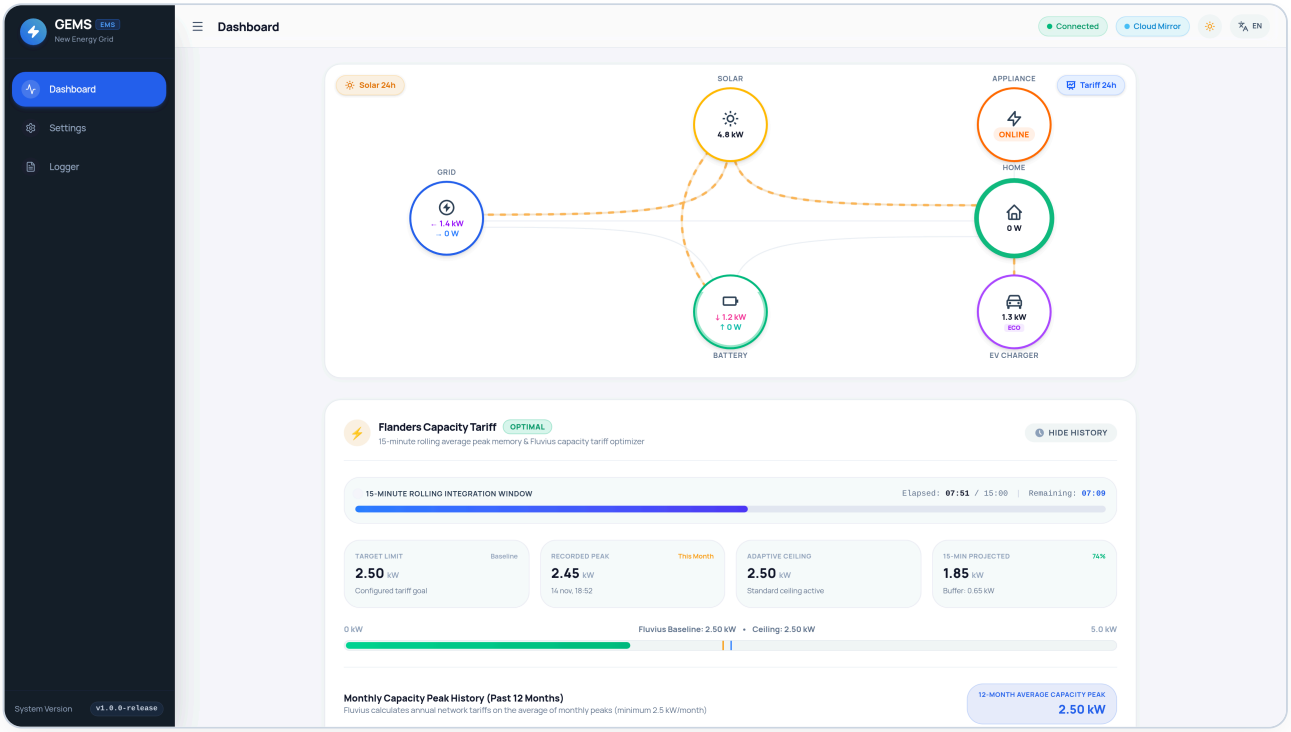
Complete electrician & homeowner manual covering **LV 51.2V parallel & HV stackable battery wiring** (`LFPWall-5000`, `LFPWall-10K-V2`, `Roller-16K`, `GroundHV-2500` & `GH02-5324`), **Enerlution App** commissioning parameters, and optimal **GRID-EMS (GEMS)** configuration.

**DOCUMENT VERSION**
v9.36.7-release (Official Tag)

**GEMS VERIFIED TEMPLATE**
enerlution_inverter (1-Watt Precision Curtailment & ESS Telemetry)

**SUPPORTED ENERLUTION INVERTERS**
L3–L12 Series • HY Series (1P LV) • HYTH-5K to HYTH-12K (3P HV)

**SUPPORTED ENERLUTION BATTERIES**
LFPWall-5000 / 10K-V2 • Roller-16K (51.2V LV) • GroundHV-2500 / GH02 (HV)




### ⚡ Why Enerlution + GEMS Deliver Seamless Local Energy Control

**Enerlution** offers both modular **51.2 V Low-Voltage** wall/mobile batteries and **High-Voltage stackable** three-phase residential/commercial towers. GEMS automatically discovers Enerlution hardware on your LAN by IEEE MAC prefix (`70:B3:D5` / `E4:5F:01`) and Modbus register signature, reads full live PV, Battery SOC/SOH, and Grid telemetry with fast/slow register tiering, executes **stepless 1-Watt solar curtailment** during negative spot prices, and coordinates EV charging and heat pumps around your battery reserve.

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Target Platform: GRID-EMS Edge Controller (Raspberry Pi ARM64)

# Section 1: Enerlution Inverter & Battery Ecosystem (LV vs. HV)

Unlike single-architecture brands, **Enerlution** manufactures two distinct voltage classes: **51.2 V Low-Voltage (LV)** single-phase storage systems and **150–600 V High-Voltage (HV)** three-phase stackable towers. Before installing or wiring any cables, confirm that your battery model matches the exact voltage class of your hybrid inverter.

> 🛑 **CRITICAL VOLTAGE SEPARATION RULE: Never Cross-Wire LV (51.2V) & HV (150–600V)!**
>
> - **Architecture A — 51.2 V Low-Voltage (`LFPWall-5000`, `LFPWall-10K-V2`, `Roller-16K`):** Wired **strictly in 51.2 V parallel** (`P+` to `P+`, `P-` to `P-`). Connect *only* to 48V/51.2V Low-Voltage hybrid inverters (`Enerlution HY / L3–L6 LV` or compatible 48V LV inverters such as `Solis -L`). **Never** wire LV wall/roller packs in series!
> - **Architecture B — High-Voltage Stackable (`GroundHV-2500`, `GH02-5324` + PDU/BCU):** Individual blocks stack **in series underneath a dedicated High-Voltage PDU / BCU Control Box** (`~153 V – 500+ Vdc`). Connect *only* to High-Voltage 3-phase hybrid inverters (`Enerlution HYTH-5K..12K` / `L5–L12 HV`). Keep the HV Manual Service Disconnect (MSD) / PDU breaker **OFF** and wear HV-insulated gloves during assembly!

## 1.1 Enerlution Battery Storage Specification Matrix

| Enerlution Battery Model | Voltage Class | Nominal Voltage & Capacity | Max Charge / Discharge | Matching Inverter Family |
|---|---|---|---|---|
| **LFPWall-5000** *(Wall-Mount $LiFePO_4$)* | **Low-Voltage** (Parallel) | `51.2 Vdc` • `102 Ah` **`5.22 kWh`** (45–57.6 V) | Charge: `40 A` (~2.05 kW) Discharge: `75 A` (~3.84 kW) | Enerlution `HY / L3–L6 LV` *(or Solis LV `-L`)* |
| **LFPWall-10K-V2** *(High-Cap Wall-Mount)* | **Low-Voltage** (Parallel) | `51.2 Vdc` • `204 Ah` **`10.44 kWh`** (45–57.6 V) | Charge: `90 A` (~4.61 kW) Discharge: `140 A` (~7.16 kW) | Enerlution `HY / L3–L6 LV` *(or Solis LV `-L`)* |
| **Roller-16K** *(Mobile Floor-Standing)* | **Low-Voltage** (Parallel) | `51.2 Vdc` • `314 Ah` **`16.08 kWh`** (45–57.6 V) | `0.5 C` heavy-duty class (Up to ~ `150 A` / ~7.6 kW) | Enerlution `HY / L5–L6 LV` *(or Solis `S6-EH1P/3P-L`)* |
| **GroundHV-2500 + PDU** *(Stackable HV Tower)* | **High-Voltage** (Series Stack) | `51.2 V` / `52 Ah` (`2.5 kWh` /blk) **`7.5 – 20+ kWh`** (3–8 blocks) | `~25 – 50 A` HV DC (`5 – 12 kW` continuous) | Enerlution **`HYTH-5K..12K`** (3-Phase HV Hybrid) |
| **GH02-5324 + BCU** *(Large HV Tower)* | **High-Voltage** (Series Stack) | High-Voltage Modular **`15.0 – 42.0 kWh`** | `~50 A` HV DC (`10 – 15+ kW` continuous) | Enerlution **`HYTH-8K..12K`** (3-Phase HV Hybrid) |

## 1.2 Enerlution Hybrid Inverter Series & Cable / Protection Sizing

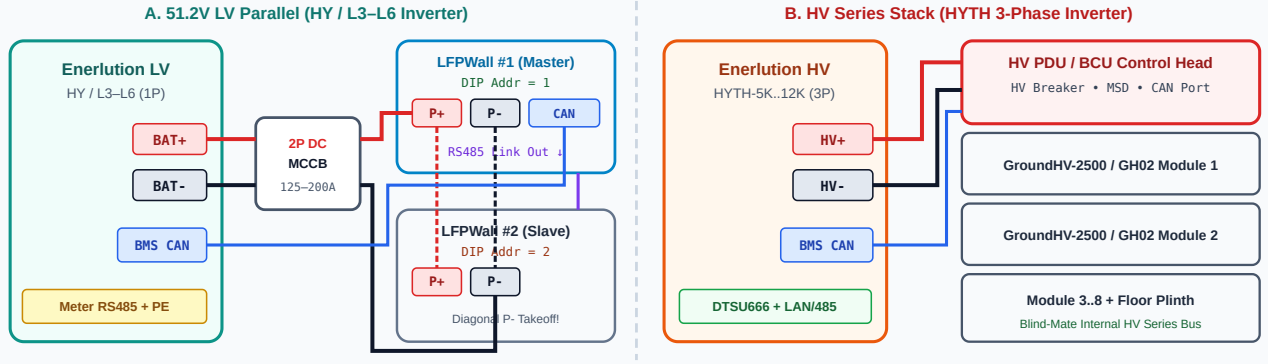
| Inverter Series | AC Grid Phases | GEMS Template | Battery DC Cable & External Protection |
|---|---|---|---|
| **Enerlution HY / L3–L6 (LV)** *(3.0 / 3.6 / 5.0 / 6.0 kW)* | Single-Phase (`230 Vac`) | `enerlution_inverter` | `25 mm²` (5 kWh) or `35–50 mm²` (10–16 kWh+) copper via **non-polarized 2-pole DC MCCB (`125 A – 200 A`, ≥80 Vdc)** |
| **Enerlution HYTH / L5–L12 (HV)** *(5.0 / 6.0 / 8.0 / 10.0 / 12.0 kW)* | Three-Phase (`3×400V+N`) | `enerlution_inverter` | Double-insulated `6 – 10 mm²` (`1000 Vdc`) HV solar/storage leads via **built-in PDU HV Breaker & Manual Service Disconnect** |

# Section 2: Step-by-Step Battery & Smart Meter Wiring Manual

De-energize the AC distribution breaker, PV DC isolator, external DC breaker / HV PDU switch, and all battery module power buttons before connecting any power or communication leads.

## 2.1 Both Wiring Topologies: LV 51.2V Parallel vs. HV Stackable Tower

**Left: Architecture A — 51.2V LV Parallel (LFPWall / Roller) | Right: Architecture B — HV Stackable (GroundHV / GH02 + PDU)**




### Architecture A — Wiring 51.2V LV Batteries (`LFPWall-5000`, `LFPWall-10K-V2`, `Roller-16K`)

1. **External 2-Pole Non-Polarized DC MCCB:** Install a non-polarized 2-pole DC breaker or `gBat` DC fuse disconnector (`≥ 80 Vdc`; `100–125 A` for a single `LFPWall-5000`, `160–200 A` for `LFPWall-10K-V2` or `Roller-16K`) between the battery bank and the inverter `BAT+ / BAT-` lugs.
2. **Parallel Balancing Rule:** For **2–3 parallel packs**, connect equal-length bridge cables (`P+` to `P+`, `P-` to `P-`) and take `BAT+` **from Pack 1 (Master)** `P+` and `BAT-` **diagonally from the final pack's** `P-`. For **4+ parallel packs**, run identical-length `25–35 mm²` fused branch leads from each battery to rated `250–400 A` tin-plated copper DC busbars.
3. **Protective Earth (`⏚ PE`):** Bond each wall/roller chassis `⏚` M6 ground terminal directly to the inverter PE bar with `≥ 6 mm²` green/yellow copper conductor.

### Architecture B — Wiring High-Voltage Towers (`GroundHV-2500` / `GH02-5324` + PDU/BCU)

1. **Mechanical Stacking & Wall Anchoring:** Level the steel base plinth against a load-bearing wall. Stack the required number of HV battery bricks vertically (minimum `3–4 × GroundHV-2500` bricks = `~153–204 Vdc` to exceed `HYTH` start voltage), fasten the anti-tip side locking brackets between every tier, and seat the **PDU / BCU High-Voltage Control Head** on top.
2. **HV DC Plugs & Safety Interlock:** Verify the PDU HV circuit breaker is **OFF** and any orange **Manual Service Disconnect (MSD)** safety plug is removed. Route double-insulated `6–10 mm²` (`1000 Vdc`) HV storage cables from the PDU `HV+ / HV-` touch-safe connectors into the `HYTH` inverter's `BAT+ / BAT-` ports until the locking collars click, and bond the base + PDU `⏚ PE` studs to the inverter earth bar.

## 2.2 BMS CAN Cable Pinout & Multi-Pack DIP Switch Addressing

Enerlution batteries establish closed-loop BMS communication (`SOC %`, `SOH %`, pack voltage, cell temperature derating, and charge/discharge current envelopes) over **CAN bus** via an RJ45 cable linking the **Master Battery / PDU** `CAN` **port** to the Enerlution inverter `BMS / CAN` **port**:

| RJ45 Pin | T568B Wire Color | Enerlution Battery / PDU Port | Inverter `BMS / CAN` Port | Wiring Notes |
|---|---|---|---|---|
| **Pin 1 & 2** | W-Orange / Orange | RS485-B / RS485-A (Optional) | RS485-B / RS485-A | Unused in CAN mode |
| **Pin 3 & 6** | W-Green / Green | Signal `GND` | Signal `GND` | Reference Ground |
| **Pin 4** | **Solid Blue** | `CAN-H` **(CAN High)** | `CAN-H` **(CAN High)** | **Mandatory CAN-H** |
| **Pin 5** | **White / Blue** | `CAN-L` **(CAN Low)** | `CAN-L` **(CAN Low)** | **Mandatory CAN-L** |
| **Pin 7 & 8** | W-Brown / Brown | Reserved / Shield GND | Reserved / Shield GND | Check factory lead label |

- **Parallel Link Cables & DIP Switches (Set Strictly While Powered OFF):**
  - **Single Battery Pack:** Keep the `ADD` DIP switches in factory standalone position (`0000` all `OFF`, or `Address 1` per enclosed V1/V2 BMS sheet).
  - **Multiple Parallel LV Packs (`LFPWall` / `Roller`):** Daisy-chain CAT5e/CAT6 link cords from **Master Pack 1** `Link Out / RS485-B` → **Slave Pack 2** `Link In / RS485-A` → **Slave Pack 3** `Link In`. Assign sequential binary addresses while powered down: **Master (Pack 1) = Address** `1` **(**`ON OFF OFF OFF`**), Slave 1 (Pack 2) = Address** `2` **(**`OFF ON OFF OFF`**), Slave 2 (Pack 3) = Address** `3` **(**`ON ON OFF OFF`**), Slave 3 (Pack 4) = Address** `4` **(**`OFF OFF ON OFF`**).** If your DIP block includes a dedicated termination bit on the last pole, set termination `ON` on the physically farthest pack.
  - **Pairing 51.2V Enerlution Packs with Third-Party LV Inverters (e.g. Solis):** Ensure the Enerlution BMS protocol matches the target inverter (`Pylon / Solis CAN` over Pins 4 & 5; switchable via DIP or Enerlution's USB-RS485 BMS Monitor utility if factory-shipped on a different OEM profile).

## 2.3 External RS-485 Smart Meter Wiring (`CHINT DTSU666` & `Eastron SDM630`)

For autonomous `Self-Consumption` battery tracking and whole-home phase visibility, wire a shielded twisted-pair RS-485 cable (`9600 bps, 8N1`, Meter Address `1`) from the main switchboard smart meter to the Enerlution inverter's `Meter / 485` terminal block:

| Smart Meter Model | Meter `RS485-A (+)` | Meter `RS485-B (-)` | Required Meter Bus Setup |
|---|---|---|---|
| **CHINT DTSU666** (3-Phase) | Terminal `24` | Terminal `25` | Address `1` • `9600 8N1` |
| **CHINT DDSU666** (1-Phase) | Terminal `5` | Terminal `6` | Address `1` • `9600 8N1` |
| **Eastron SDM630-Modbus V2** | Terminal `A / 485A` | Terminal `B / 485B` | Address `1` • `9600 8N1` |

*CT Orientation Check:* Clamp all CTs on the main incoming utility phase conductors immediately after the utility meter (upstream of all house breakers, EV wallboxes, and solar/storage inverters) with CT arrows pointing strictly **Grid → Home/Load**.

## 2.4 Safe Soft-Start Power-Up & Shutdown Sequence

⚡ **Mandatory Pre-Charge Startup Order**

1. **Verify Polarity Cold:** With all breakers OFF, verify `BAT+ / BAT-` polarity with a DC voltmeter.
2. **Close DC Isolator First:** Switch the LV 2-pole DC MCCB **ON** (or insert the HV MSD safety plug and switch the PDU HV breaker **ON**) while the BMS is still asleep.
3. **Wake Master / PDU BMS:** Press and hold the Master battery or HV PDU `POWER / SW` **button (** `2–3 seconds` **)** so the internal pre-charge circuit softly energizes the inverter capacitors. Confirm green `RUN` LED is solid/blinking normally and red `ALM` LED is OFF.
4. **Energize PV & Grid:** Switch the inverter **PV DC isolator ON**, then turn **ON** the **AC Grid breaker**.

*Shutdown (Reverse):* **AC Breaker OFF** → **PV DC Switch OFF** → Hold Master/PDU `POWER` button 3s to sleep BMS → **DC MCCB / PDU Breaker OFF** (wait 5 minutes before touching HV terminals!).

# Section 3: Step-by-Step Commissioning in the Enerlution App

Download the official **Enerlution** mobile app (iOS App Store / Google Play) to commission Wi-Fi/LAN communication, confirm BMS handshakes, select smart meter profiles, and leave active power limits unlocked for GEMS.

## 3.1 Connecting Locally via Bluetooth or Wi-Fi Access Point

1. Plug the **Enerlution Wi-Fi / LAN communication module** firmly into the bottom `COM / Wi-Fi` port of the inverter.
2. Enable **Bluetooth** and **Wi-Fi** on your smartphone (**temporarily turn Mobile Data OFF** if joining the module's local Wi-Fi AP SSID matching the logger serial number so your phone does not drop offline traffic).
3. Open the **Enerlution App**, select **Local Commissioning / Device Debugging**, log in with your **Installer** credentials, and bind the communication stick to your local **2.4 GHz Wi-Fi** or **wired Ethernet LAN** (reserve a **static DHCP IPv4 address** on your router!).

## 3.2 Complete Enerlution App Parameter Checklist for GEMS Compatibility

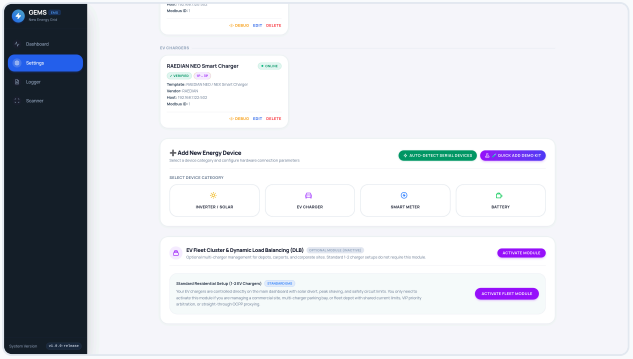
| Enerlution App Setting | Required Value | Why This Exact Setting Is Critical for GEMS |
|---|---|---|
| **Battery Type & Protocol** *(Battery Settings)* | **Lithium-Ion (CAN)** *(Enerlution / Pylon Profile)* | Activates closed-loop BMS telemetry (`SOC %`, `SOH %`, pack voltage, live current & thermal protection). Confirm battery state shows **Normal / Standby / Charging** (`State 1–5`, never `7 Offline`). |
| **EMS Working Mode** *(Energy Management Mode)* | **Self-Consumption** **NOT Full Off-Grid / Static Force** | Instructs the Enerlution inverter to autonomously store excess daytime PV into your `LFPWall` or `GroundHV` stack and discharge seamlessly to cover household loads and evening peaks! |
| **Minimum / Reserve SOC** *(End-of-Discharge SOC)* | **10 %** *(Standard Grid-Tied)* **15 % – 20 %** *(With EPS Backup)* | Protects LiFePO$_4$ cell longevity (90% DoD) while reserving blackout backup capacity if critical loads are wired to the `EPS / Backup` port. Keep **Max Charge SOC = 100%** for cell balancing. |
| **External Smart Meter** *(Meter / CT Settings)* | **Enable** → `CHINT DTSU666` or `SDM630` | Enables sub-second behind-the-meter export/import sensing on the inverter (`Address 1`, `9600 bps`) and feeds live meter power into GEMS. Run a 2 kW kettle check in the app to verify positive import polarity! |
| **Active Power Control / Export Limit** | **Disabled / Unlimited (0)** *(Unless legally mandated by DSO)* | **Crucial!** Leaving static zero-export *Disabled* in the Enerlution App allows profitable solar export and Green Certificates, while letting GEMS dynamically write **1-Watt precision curtailment limits** (`0 W` to `Rated W`) whenever spot prices turn negative! |
| **Modbus Communication** *(RS-485 & TCP Server)* | Slave ID: **1** • Baud: **9600 8N1** Modbus TCP Port: **502** | Matches GEMS's native `enerlution_inverter` driver defaults over both wired LAN/Wi-Fi (`TCP :502`) and direct USB-to-RS485 (`9600 8N1`). |

# Section 4: Onboarding & Optimizing Enerlution in GEMS

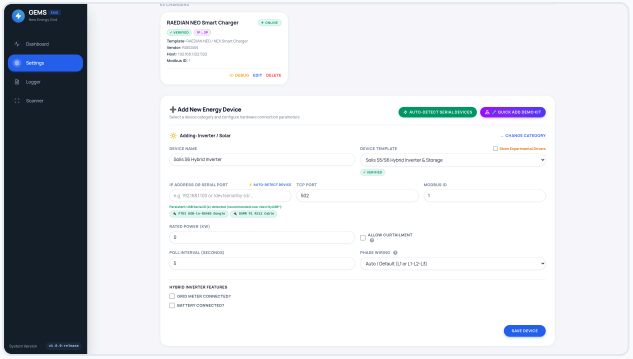
Open the GEMS web UI (`https://gems.local` or appliance IP), navigate to **Settings**, click the **padlock icon** 🔒 in the top-right header, and enter your **Installer Access PIN** to enable **Certified Installer Mode**.

## 4.1 Automatic Discovery & Onboarding (`Settings → Devices`)

GEMS includes native IEEE MAC OUI recognition (`70:B3:D5` and `E4:5F:01`) and ASCII model/serial register fingerprinting for Enerlution—clicking **Scan Subnet** in the **Scanner** tab automatically detects networked Enerlution units on port `502`! You can also connect via direct **USB-to-RS485** (`/dev/serial/by-id/...`, `9600 bps, 8N1`, Slave ID `1`) for zero-Wi-Fi industrial reliability:



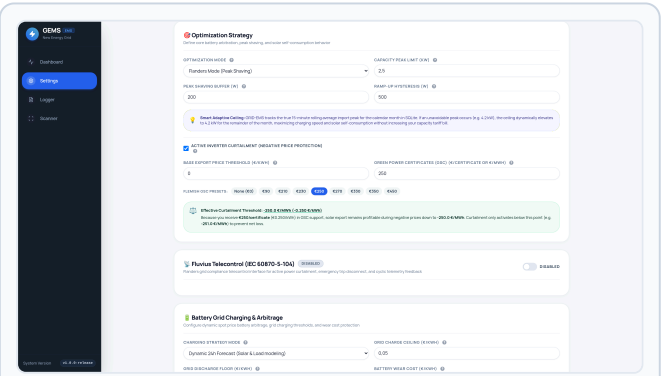
1. Select Smart Meter (P1) First, Then Inverter / Solar



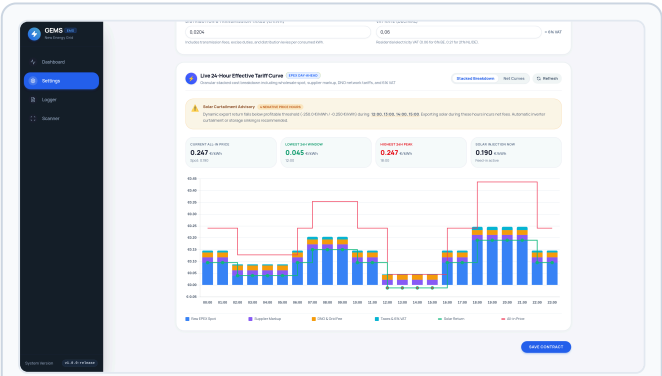
2. Configure Enerlution Endpoint, Rated kW, Curtailment & Battery kWh

| GEMS Device Field | Recommended Value | Why This Setting Matters |
|---|---|---|
| **Primary Grid Meter** | `P1 Serial Meter` (`p1_serial`) | Onboard your digital utility P1 meter first! Gives GEMS official 3-phase currents and quarter-hour Flanders capacity peak billing accuracy alongside Enerlution's internal meter. |
| **Device Template** | `Enerlution Hybrid Inverter` (`enerlution_inverter`) | Activates 80 ms bus pacing, fast/slow 5-minute metadata caching, SOH/SOC telemetry, and **1-Watt precision Fixed Active Power curtailment**. *(Note: If an Enerlution 51.2V battery is wired to a Solis LV inverter instead, select the Solis template!).* |
| **IP Address or Serial Port** | Reserved Static IP (e.g. `192.168.1.130`) or `/dev/serial/by-id/...` | Use wired Ethernet or USB-RS485 whenever possible; if using Wi-Fi, ensure strong 2.4 GHz RSSI (`> -65 dBm`) and a router DHCP reservation. |
| **TCP Port & Modbus ID** | Port: `502` • Modbus ID: `1` | Matches standard Enerlution Modbus TCP and RS-485 Slave ID `1`. |
| **Rated Power (kW)** | Exact Inverter AC Nameplate kW *(e.g. `3.6`, `5.0`, `6.0`, `8.0`, `10.0`, `12.0`)* | **Do not leave at 0!** Bounds GEMS's 1-Watt active power curtailment ceiling accurately to your physical inverter's AC capacity. |
| **Allow Curtailment** | **Checked (✓)** | Permits GEMS to write exact Watt limits during negative EPEX spot prices or zero-export operation and release back to unrestricted output automatically. |
| **Poll Interval (seconds)** | **3 s** *(RS-485 / Wired LAN)* **5 s** *(2.4 GHz Wi-Fi Stick)* | GEMS caches static ASCII/lifetime registers for 5 minutes so every fast poll reads only compact real-time power and SOC registers! |
| **Phase Wiring** | `L1`, `L2`, or `L3` (1P `HY / L3–L6`) or `L1-L2-L3` (3P `HYTH`) | Identifies the exact breaker phase on single-phase installations for accurate per-phase main fuse protection. |
| **Grid Meter & Battery Flags** | Grid Meter: **Checked (✓)** if DTSU/SDM wired Battery: **Checked (✓)** + Exact kWh (`5.2`, `10.4`, `16.1`, `10.0`, `15.0`...) | Displays live `LFPWall` / `GroundHV` SOC % and charge/discharge watts on the PowerFlow canvas and synchronizes EV solar surplus balancing with battery state! |

# Section 5: Best-Practice GEMS Strategy, EV & Tariff Coordination



Flanders Peak Shaving, Netherlands Mode & GSC Subsidy Shield



24-Hour Dynamic Import & Solar Injection Price Curve

## 5.1 Regional Optimization Recipes (`Settings → Strategy`)

### 🇧🇪 Recipe A: Flanders (Peak Shaving)

- **Mode:** `Flanders Mode (Peak Shaving)`
- **Capacity Peak Limit:** `2.5 kW` (standard) or `3.0–3.5 kW` (heat pump + EV).
- **Peak Shaving Buffer:** `200 W`
- **Active Curtailment:** `Checked (✓)`, Base `0.00 €/kWh`
- **GSC Subsidy Shield:** Enter `€0` (modern PV) or exact certificate value (`€250–€450/MWh`). GEMS automatically keeps Enerlution solar exporting down to `-GSC/1000 €/kWh`!
- **How It Works:** Enerlution's `Self-Consumption` loop discharges the battery to cover cooking/heat-pump spikes while GEMS throttles EV charging before any 15-min billing peak is breached.

### 🇳🇱 Recipe B: Netherlands (Saldering)

- **Mode:** `Netherlands Mode (Smart Saldering)`
- **Active Curtailment:** `Checked (✓)`
- **Base Export Price Threshold:** `0.00 €/kWh` (or your supplier's feed-in levy threshold).
- **How It Works:** Daytime PV surplus fills your Enerlution `LFPWall` / `GroundHV` stack first, then ramps **Raedian** EV charging (`6A–32A`) and SG-Ready hot water. Once storage is 100% full during negative-return hours, GEMS regulates Enerlution output in **1-Watt steps** so grid export drops cleanly to `0 W`!

### 🌿 Recipe C: Fixed / Dual-Tariff Eco

- **Mode:** `Eco (Max Self-Consumption)`
- **Active Curtailment:** `Unchecked (□)` (unless on a contract with negative export penalties).
- **Grid Fuse Protection:** Enter exact main breaker amps (`25A / 32A / 40A`) and `Meter Offline Fallback = 0 A`.
- **How It Works:** Pure autonomous solar self-consumption: Enerlution prioritizes battery charging, GEMS steers surplus solar into EVs and thermal relays, and main fuses are protected 24/7.

## 5.2 Coordinating Enerlution Storage with EV Chargers (Raedian Wallboxes)

- **Daytime Solar Surplus Priority:** Set your **Raedian NEO / NEX / Gemini** wallbox to `PV Only (Solar)` or `Eco` on the GEMS Dashboard. Because Enerlution's behind-the-meter loop absorbs initial PV surplus into the battery first, GEMS automatically waits until excess solar exceeds your configured EV threshold (e.g. `1.4 kW` 1-phase) before ramping EV charging—preventing the EV charger from stealing solar away from an empty home battery or seesawing against it!
- **Smart Cheapest-Hours EV Charging:** Enable **Smart EV Cheapest Hours** (`Settings → Strategy`, e.g. `3–4 cheapest hours`) alongside your **Energy Contract** formula (`Settings → Energy`

`Contract`). If you schedule high-power overnight EV charging and want to preserve your Enerlution battery for the morning household peak, either schedule EV charging during windows when the Enerlution App's optional *Discharge Window* is paused or cap EV charging current below your Flanders capacity ceiling.

# Section 6: Verification Checklist & Troubleshooting Matrix

## 6.1 6-Point Post-Commissioning Handover Checklist

✅ **Mandatory Field Verification Before Leaving Site**

1. **BMS & SOH Verification:** Confirm zero red `ALM` LEDs on all `LFPWall`, `Roller`, or `GroundHV` modules and verify live `SOC %` and `SOH %` appear in the Enerlution App (`Battery State = Charging / Discharging / Standby`).
2. **GEMS Online & Debug Modal:** In **Settings → Connected Devices**, confirm `Enerlution Hybrid Inverter` displays `● Online`. Click **Debug** and verify ~230 V grid voltage, plausible PV watts, and live Battery SOC %.
3. **1-Watt Curtailment Release Check:** Verify in the **Logger** tab that during normal positive-price operation GEMS logs `Released active power limit (40400=0)` with zero write errors.
4. **Kettle Load Polarity Check:** Boil a ~2 kW kettle and confirm Home Consumption increases by ~2 kW on the GEMS Dashboard while the Enerlution battery smoothly ramps discharge to cover the load.
5. **Export Configuration Backup:** In **Settings → System Info**, click **Export Configuration Backup (.json)**.
6. **Lock Certified Installer Mode:** Click the padlock icon 🔒 in the Settings header to lock critical device and breaker settings.

## 6.2 Symptom Resolution & Troubleshooting Matrix

| Observed Symptom | Probable Root Cause | Field Corrective Action |
|---|---|---|
| Battery State reads `Offline (7)` or `Fault (6)` | CAN cable plugged into `Link/RS485` port instead of `CAN`, wrong BMS protocol, or DIP switches changed while ON. | Power all modules OFF, set unique `ADD` DIP switches, connect RJ45 (`Pin 4 CAN-H`, `Pin 5 CAN-L`) strictly between Master/PDU `CAN` and Inverter `BMS`, and select `Lithium-Ion (CAN)` in the Enerlution App. |
| HV `HYTH` inverter does not recognize `GroundHV-2500` tower | Fewer than minimum series HV blocks installed, HV MSD plug unseated, or PDU breaker OFF. | Verify minimum brick count for inverter start voltage (`≥ 3–4 blocks`), confirm orange MSD plug is locked, switch PDU HV breaker ON, then wake PDU `POWER` button. |
| GEMS reports `Connection refused` or `i/o timeout` on `:502` | Wi-Fi stick dropped off 5 GHz band, IP changed via DHCP, or wrong Slave ID. | Bind stick strictly to a **2.4 GHz Wi-Fi** SSID (or use wired LAN / direct USB-RS485 `9600 8N1`), assign a router DHCP reservation, and verify `Modbus ID = 1`. |
| Inverter caps PV output even when spot prices are positive | Static Zero-Export / Active Power Limit was left enabled inside the Enerlution App. | Open Enerlution App → `Active Power / Export Control` and set to `Disabled / Unlimited`, allowing GEMS to dynamically release (`40400=0`) and engage (`40400=1`) curtailment. |
| Battery charges from AC when house load turns ON | `DTSU666` / `SDM630` CT clamps reversed or `RS485-A / B` / phase voltage taps mismatched. | Ensure CT arrows point **Grid → Home/Load**, verify `DTSU666` Pins `24 (A+)` & `25 (B-)`, and confirm `L1/L2/L3` voltage taps match `CT1/CT2/CT3`. |
| Curtailment caps at wrong wattage | `Rated Power (kW)` left inaccurate on the GEMS device card. | Edit the Enerlution device in **Settings → Devices** and enter the exact inverter AC nameplate kW (e.g. `5.0`, `8.0`, `10.0`, `12.0`). |

## 6.3 Live Diagnostic Log Verification

Installers can verify live Enerlution telemetry and 1-Watt curtailment writes in the GEMS **Logger** tab or Cockpit Terminal (`:9090`):

```
# Inspect live Enerlution polling and active power limit events via Cockpit Terminal:
sudo journalctl -u gems.service -f | grep -i "Enerlution"

# Expected healthy output during connection, negative-price curtailment, and profitable release:
[INFO] EnerlutionPoller [Enerlution HYTH-10K]: Connected via tcp://192.168.1.130:502 (ID: 1)
[INFO] EnerlutionPoller [Enerlution HYTH-10K]: Set active power limit to 1450 W
[INFO] EnerlutionPoller [Enerlution HYTH-10K]: Released active power limit (40400=0)
```

> 📞 **Fleet Monitoring & Installer Support**
> Pair your GEMS unit with **GRID-EMS-SERVER** (`https://ems.newenergygrid.com`) under **Settings → Cloud Server** for remote 24-hour forecast diagnostics, live telemetry verification, and centralized fleet management by **New Energy Grid** & **The Charge Grid**.